25901227 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCECT203 - WIRELESS TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which modulation technique provides better noise immunity? [U] / [CO1]

   a). Amplitude b). FM

   c). PM d). ASK

2. RF systems are widely used in: [U] / [CO1]

   a). Washing machines b). Wireless communication

   c). Printers d). Calculators

3. The radio frequency range typically lies between: [U] / [CO1]

   a). 0 Hz – 1 kHz b). 1 kHz – 1 MHz

   c). 3 kHz – 300 GHz d). Above 300 GHz

4. Fading in wireless channels is mainly caused by: [U] / [CO2]

   a). Thermal noise b). Bandwidth limitation

c). Multipath propagation and mobility

d). Modulation technique

5. Which type of fading occurs due to large obstacles like buildings or hills? [U] / [CO2]

a). Fast fading

b). Small-scale fading

c). Shadow fading

d). Flat fading

6. SNR stands for [U] / [CO2]

a). Signal Noise Ratio

b). System Noise Ratio

c). Signal-to-Noise Ratio

d). Signal Network Ratio

7. Which mixer type provides better isolation and reduced noise? [U] / [CO3]

a). Single-ended mixer

b). Balanced mixer

c). Passive mixer

d). Unbalanced mixer

8. A basic RF transmitter consists of [K] / [CO3]

a). Modulator, amplifier, antenna

b). Only antenna

c). Only oscillator

d). Only filter

9. Wi-Fi is based on which IEEE standard? [U] / [CO3]

a). IEEE 802.3

b). IEEE 802.11

c). IEEE 802.15

d). IEEE 802.16

10. LNA mainly improves [U] / [CO3]

a). Bandwidth

b). Power efficiency

c). Signal-to-Noise Ratio (SNR)

d). Carrier frequency

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. What is a carrier signal? [U] / [CO1]

12. Define RF spectrum. [K] / [CO1]

13. A wireless link has signal power 10 mW and interference power 2 mW. Calculate SIR. [A] / [CO2]

14. Define link budget. [K] / [CO2]

15. Explain the radiation pattern of an antenna in detail. [U] / [CO3]

16. List basic types of RF filters. [K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Compare licensed and unlicensed spectrum bands. Discuss their advantages, disadvantages and applications. [U] / [CO1]

18. Explain the bandwidth–data rate trade-off in modern wireless systems using higher order modulation schemes. [A] / [CO1]

19. Differentiate between dB and dBm. [U] / [CO2]

20. Compare LC oscillators and crystal oscillators. [U] / [CO3]

[10x2 = 20]